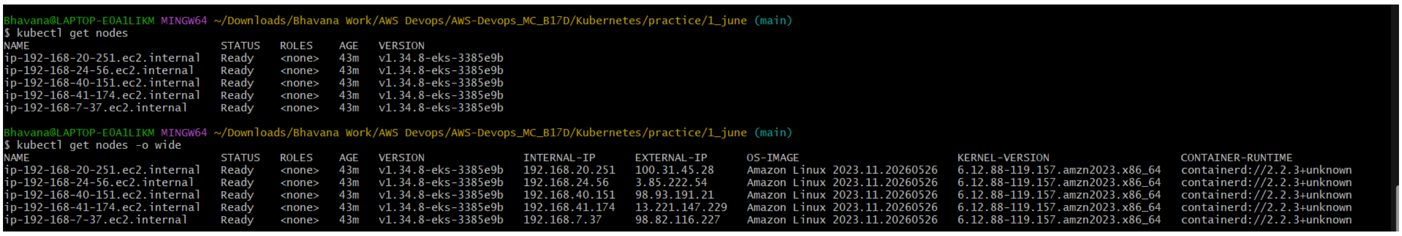
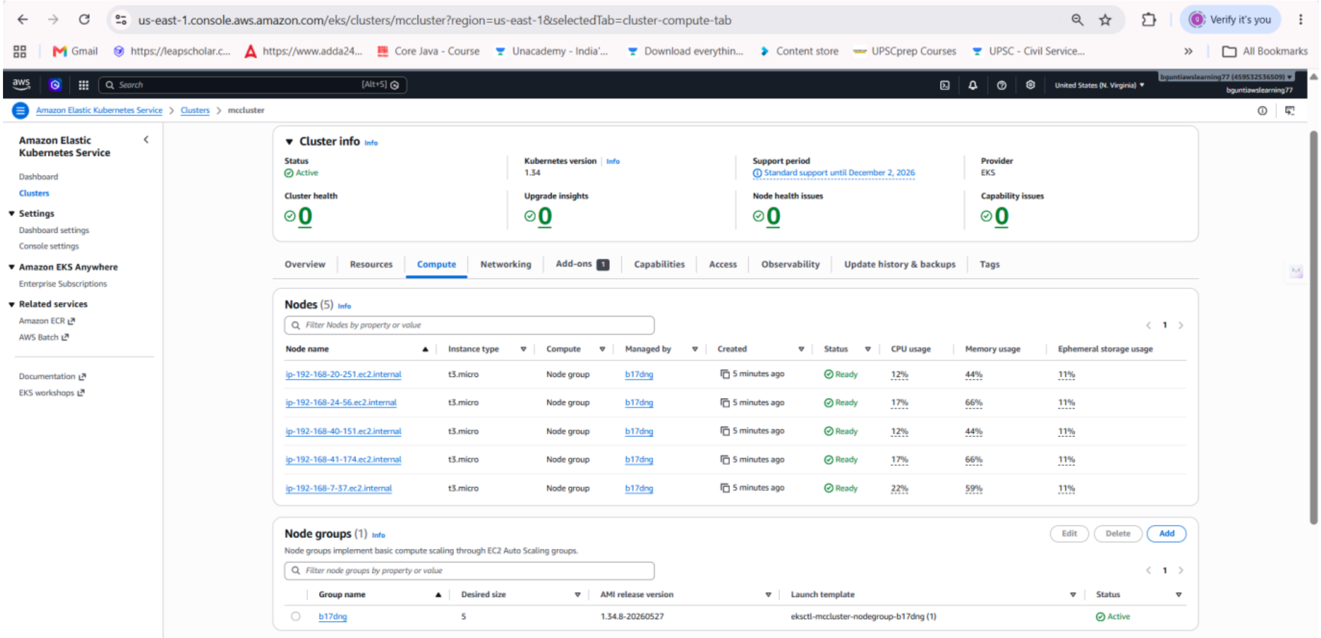
Kubernetes Day 4

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ eksctl version
0.226.0

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ kubectl version --client
Client Version: v1.35.3-eks-bbe087e
Kustomize Version: v5.7.1

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ aws --version
aws-cli/2.34.53 Python/3.14.5 Windows/11 exe/AMD64

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ eksctl create cluster --name mccluster --nodegroup-name b17dng --node-type t3.micro --nodes 5 –managed

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ eksctl get cluster
NAME            REGION          EKSCTL CREATED
mccluster       us-east-1       True


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ kubectl get nodes
NAME                             STATUS   ROLES    AGE    VERSION
ip-192-168-20-251.ec2.internal   Ready    <none>   109s   v1.34.8-eks-3385e9b
ip-192-168-24-56.ec2.internal    Ready    <none>   113s   v1.34.8-eks-3385e9b
ip-192-168-40-151.ec2.internal   Ready    <none>   108s   v1.34.8-eks-3385e9b
ip-192-168-41-174.ec2.internal   Ready    <none>   104s   v1.34.8-eks-3385e9b
ip-192-168-7-37.ec2.internal     Ready    <none>   113s   v1.34.8-eks-3385e9b
```



```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get nodes
NAME                             STATUS   ROLES    AGE   VERSION
ip-192-168-20-251.ec2.internal   Ready    <none>   43m   v1.34.8-eks-3385e9b
ip-192-168-24-56.ec2.internal    Ready    <none>   43m   v1.34.8-eks-3385e9b
ip-192-168-40-151.ec2.internal   Ready    <none>   43m   v1.34.8-eks-3385e9b
ip-192-168-41-174.ec2.internal   Ready    <none>   43m   v1.34.8-eks-3385e9b
ip-192-168-7-37.ec2.internal     Ready    <none>   43m   v1.34.8-eks-3385e9b

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get nodes -o wide
NAME                             STATUS   ROLES    AGE   VERSION               INTERNAL-IP      EXTERNAL-IP      OS-IMAGE                         KERNEL-VERSION                        CONTAINER-RUNTIME
ip-192-168-20-251.ec2.internal   Ready    <none>   43m   v1.34.8-eks-3385e9b   192.168.20.251   100.31.45.28     Amazon Linux 2023.11.20260526    6.12.88-119.157.amzn2023.x86_64       containerd://2.2.3+unknown
ip-192-168-24-56.ec2.internal    Ready    <none>   43m   v1.34.8-eks-3385e9b   192.168.24.56    3.85.222.54      Amazon Linux 2023.11.20260526    6.12.88-119.157.amzn2023.x86_64       containerd://2.2.3+unknown
ip-192-168-40-151.ec2.internal   Ready    <none>   43m   v1.34.8-eks-3385e9b   192.168.40.151   98.93.191.21     Amazon Linux 2023.11.20260526    6.12.88-119.157.amzn2023.x86_64       containerd://2.2.3+unknown
ip-192-168-41-174.ec2.internal   Ready    <none>   43m   v1.34.8-eks-3385e9b   192.168.41.174   13.221.147.229   Amazon Linux 2023.11.20260526    6.12.88-119.157.amzn2023.x86_64       containerd://2.2.3+unknown
ip-192-168-7-37.ec2.internal     Ready    <none>   43m   v1.34.8-eks-3385e9b   192.168.7.37     98.82.116.227    Amazon Linux 2023.11.20260526    6.12.88-119.157.amzn2023.x86_64       containerd://2.2.3+unknown
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ kubectl get pods
No resources found in default namespace.
```




```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ kubectl get all
NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   11m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ kubectl api-resources

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ kubectl run pod1 --image tomcat
pod/pod1 created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ kubectl get all
NAME       READY   STATUS              RESTARTS   AGE
pod/pod1   0/1     ContainerCreating   0          8s

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   12m
```

1_pod.yaml

```yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: fb-pod #pod name
  labels:
    app: fb-app # this is the label of the pod, we can use this label to select the pod.
    env: prod
spec:
  containers:
  - name: fb-container #container name used in the pod
    image: tomcat
    ports:
    - containerPort: 8080 # tomcat server port.
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
$ cd practice/

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
$ ll
total 0
drwxr-xr-x 1 Bhavana 197121 0 Jun  2 11:40 1_june/
drwxr-xr-x 1 Bhavana 197121 0 Jun  2 11:39 29_May/

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
$ cd 1_june/

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ cat 1_pod.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: fb-pod #pod name
  labels:
    app: fb-app # this is the label of the pod, we can use this label to select the pod.
    env: prod
spec:
  containers:
  - name: fb-container #container name used in the pod
    image: tomcat
    ports:
    - containerPort: 8080 # tomcat server port.


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl apply -f 1_pod.yaml
pod/fb-pod created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get po
NAME     READY   STATUS    RESTARTS   AGE
fb-pod   1/1     Running   0          12s
pod1     1/1     Running   0          10m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl describe pod fb-pod
Name:             fb-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             ip-192-168-40-151.ec2.internal/192.168.40.151
Start Time:       Tue, 02 Jun 2026 11:49:16 +0530
Labels:           app=fb-app
                  env=prod
Annotations:      <none>
Status:           Running
IP:               192.168.36.105
IPs:
  IP:  192.168.36.105
Containers:
  fb-container:
    Container ID:
containerd://24272cf1687b41c3ee85b5478a64439a896f74fb98b992f90be796cc6d5e4d76
    Image:          tomcat
```

```
    Image ID:
docker.io/library/tomcat@sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea
0e5071e19facc1e5e7
    Port:          8080/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Tue, 02 Jun 2026 11:49:23 +0530
    Ready:         True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-ptxhz
(ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-ptxhz:
    Type:                    Projected (a volume that contains injected data
from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists
for 300s
                             node.kubernetes.io/unreachable:NoExecute
op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  30s   default-scheduler  Successfully assigned default/fb-pod to
ip-192-168-40-151.ec2.internal
  Normal  Pulling    29s   kubelet            spec.containers{fb-container}: Pulling
image "tomcat"
  Normal  Pulled     23s   kubelet            spec.containers{fb-container}:
Successfully pulled image "tomcat" in 6.088s (6.088s including waiting). Image size:
154490448 bytes.
  Normal  Created    23s   kubelet            spec.containers{fb-container}: Created
container: fb-container
  Normal  Started    23s   kubelet            spec.containers{fb-container}: Started
container fb-container


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl run pod2 --image tomcat --dry-run -o yaml
W0602 11:56:51.275545   25840 helpers.go:719] --dry-run is deprecated and can
be replaced with --dry-run=client.
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: pod2
  name: pod2
spec:
  containers:
  - image: tomcat
    name: pod2
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
```

2_pod.yaml

```yaml
# Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
# $ kubectl run pod2 --image tomcat --dry-run -o yaml
# W0602 11:56:51.275545   25840 helpers.go:719] --dry-run is deprecated and can be replaced with --dry-run=client.
apiVersion: v1
kind: Pod
metadata:
  name: pod2
  labels:
    run: pod2

spec:
  containers:
  - name: pod2
    image: tomcat
    resources: {}
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get nodes
NAME                            STATUS   ROLES    AGE   VERSION
ip-192-168-20-251.ec2.internal  Ready    <none>   28m   v1.34.8-eks-3385e9b
ip-192-168-24-56.ec2.internal   Ready    <none>   28m   v1.34.8-eks-3385e9b
ip-192-168-40-151.ec2.internal  Ready    <none>   28m   v1.34.8-eks-3385e9b
ip-192-168-41-174.ec2.internal  Ready    <none>   28m   v1.34.8-eks-3385e9b
ip-192-168-7-37.ec2.internal    Ready    <none>   28m   v1.34.8-eks-3385e9b

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods
NAME     READY   STATUS    RESTARTS   AGE
fb-pod   1/1     Running   0          14m
pod1     1/1     Running   0          24m
```

```yaml
#3_multicont_pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: two-container-pod
  labels:
    app: two-container-app
spec:
  containers:
  - name: container1
    image: tomcat
    ports:
    - containerPort: 8080
  - name: container2
    image: busybox
    command: ['sh', '-c', 'echo "Hello, Kubernetes!" && sleep 3600']
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
```

```
$ kubectl apply -f 3_multicont_pod.yaml
pod/two-container-pod created


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
fb-pod              1/1     Running   0          14m
pod1                1/1     Running   0          25m
two-container-pod   2/2     Running   0          14s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl describe pod two-container-pod

Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  2m47s  default-scheduler  Successfully assigned default/two-container-pod to ip-192-168-41-174.ec2.internal
  Normal  Pulling    2m47s  kubelet            spec.containers{container1}: Pulling image "tomcat"
  Normal  Pulled     2m41s  kubelet            spec.containers{container1}: Successfully pulled image "tomcat" in 5.785s (5.787s including waiting). Image size: 154490448 bytes.
  Normal  Created    2m41s  kubelet            spec.containers{container1}: Created container: container1
  Normal  Started    2m41s  kubelet            spec.containers{container1}: Started container container1
  Normal  Pulling    2m41s  kubelet            spec.containers{container2}: Pulling image "busybox"
  Normal  Pulled     2m40s  kubelet            spec.containers{container2}: Successfully pulled image "busybox" in 649ms (650ms including waiting). Image size: 2236931 bytes.
  Normal  Created    2m40s  kubelet            spec.containers{container2}: Created container: container2
  Normal  Started    2m40s  kubelet            spec.containers{container2}: Started container container2


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl logs two-container-pod -c container1
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl logs two-container-pod -c container1
02-Jun-2026 06:33:58.638 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Server version name:   Apache Tomcat/11.0.22
02-Jun-2026 06:33:58.648 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Server built:          May 1 2026 18:29:05 UTC
02-Jun-2026 06:33:58.648 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Server version number: 11.0.22.0
02-Jun-2026 06:33:58.648 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log OS Name:               Linux
02-Jun-2026 06:33:58.648 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log OS Version:            6.12.88-119.157.amzn2023.x86_64
02-Jun-2026 06:33:58.648 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Architecture:          amd64
02-Jun-2026 06:33:58.649 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Java Home:             /opt/java/openjdk
02-Jun-2026 06:33:58.649 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log JVM Version:           25.0.3+9-LTS
02-Jun-2026 06:33:58.649 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log JVM Vendor:            Eclipse Adoptium
02-Jun-2026 06:33:58.649 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log CATALINA_BASE:         /usr/local/tomcat
02-Jun-2026 06:33:58.649 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log CATALINA_HOME:         /usr/local/tomcat
02-Jun-2026 06:33:58.674 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Djava.util.logging.config.file=/usr/local/tomcat/conf/logging.properties
02-Jun-2026 06:33:58.675 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager
02-Jun-2026 06:33:58.675 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Djdk.tls.ephemeralDHKeySize=2048
02-Jun-2026 06:33:58.675 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Dorg.apache.catalina.security.SecurityListener.UMASK=0027
02-Jun-2026 06:33:58.675 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: --add-opens=java.base/java.lang=ALL-UNNAMED
02-Jun-2026 06:33:58.675 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: --add-opens=java.base/java.lang.reflect=ALL-UNNAMED
02-Jun-2026 06:33:58.675 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: --add-opens=java.base/java.io=ALL-UNNAMED
02-Jun-2026 06:33:58.675 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: --add-opens=java.base/java.util=ALL-UNNAMED
02-Jun-2026 06:33:58.676 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: --add-opens=java.base/java.util.concurrent=ALL-UNNAMED
02-Jun-2026 06:33:58.676 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
02-Jun-2026 06:33:58.676 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: --enable-native-access=ALL-UNNAMED
02-Jun-2026 06:33:58.676 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Dcatalina.base=/usr/local/tomcat
02-Jun-2026 06:33:58.676 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Dcatalina.home=/usr/local/tomcat
02-Jun-2026 06:33:58.676 INFO [main] org.apache.catalina.startup.VersionLoggerListener.log Command line argument: -Djava.io.tmpdir=/usr/local/tomcat/temp
02-Jun-2026 06:33:58.698 INFO [main] org.apache.catalina.core.AprLifecycleListener.lifecycleEvent Loaded Apache Tomcat Native library [2.0.14] using APR version [1.7.2].
02-Jun-2026 06:33:58.727 INFO [main] org.apache.catalina.core.AprLifecycleListener.initializeSSL OpenSSL successfully initialized [OpenSSL 3.0.13 30 Jan 2024]
02-Jun-2026 06:33:59.331 INFO [main] org.apache.coyote.AbstractProtocol.init Initializing ProtocolHandler ["http-nio-8080"]
02-Jun-2026 06:33:59.387 INFO [main] org.apache.catalina.startup.Catalina.load Server initialization in [1233] milliseconds
02-Jun-2026 06:33:59.473 INFO [main] org.apache.catalina.core.StandardService.startInternal Starting service [Catalina]
02-Jun-2026 06:33:59.473 INFO [main] org.apache.catalina.core.StandardEngine.startInternal Starting Servlet engine: [Apache Tomcat/11.0.22]
02-Jun-2026 06:33:59.494 INFO [main] org.apache.coyote.AbstractProtocol.start Starting ProtocolHandler ["http-nio-8080"]
02-Jun-2026 06:33:59.516 INFO [main] org.apache.catalina.startup.Catalina.start Server startup in [124] milliseconds
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl logs two-container-pod -c container2
Hello, Kubernetes!
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods -o wide
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP               NODE                             NOMINATED NODE   READINESS GATES
fb-pod              1/1     Running   0          26m   192.168.36.105   ip-192-168-40-151.ec2.internal   <none>           <none>
pod1                1/1     Running   0          37m   192.168.25.228   ip-192-168-20-251.ec2.internal   <none>           <none>
two-container-pod   2/2     Running   0          12m   192.168.58.252   ip-192-168-41-174.ec2.internal   <none>           <none>
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl cordon ip-192-168-40-151.ec2.internal
node/ip-192-168-40-151.ec2.internal cordoned

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get nodes
NAME                             STATUS                     ROLES    AGE
VERSION
ip-192-168-20-251.ec2.internal   Ready                      <none>   46m
v1.34.8-eks-3385e9b
ip-192-168-24-56.ec2.internal    Ready                      <none>   46m
v1.34.8-eks-3385e9b
ip-192-168-40-151.ec2.internal   Ready,SchedulingDisabled   <none>   46m
v1.34.8-eks-3385e9b
ip-192-168-41-174.ec2.internal   Ready                      <none>   46m
v1.34.8-eks-3385e9b
ip-192-168-7-37.ec2.internal     Ready                      <none>   46m
v1.34.8-eks-3385e9b

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP               NODE
NOMINATED NODE   READINESS GATES
fb-pod              1/1     Running   0          29m   192.168.36.105   ip-
192-168-40-151.ec2.internal   <none>           <none>
pod1                1/1     Running   0          39m   192.168.25.228   ip-
192-168-20-251.ec2.internal   <none>           <none>
two-container-pod   2/2     Running   0          14m   192.168.58.252   ip-
192-168-41-174.ec2.internal   <none>           <none>

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl uncordon ip-192-168-40-151.ec2.internal
node/ip-192-168-40-151.ec2.internal uncordoned

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get nodes -o wide
NAME                             STATUS   ROLES    AGE   VERSION
INTERNAL-IP      EXTERNAL-IP      OS-IMAGE                        KERNEL-
VERSION
       CONTAINER-RUNTIME
ip-192-168-20-251.ec2.internal   Ready    <none>   48m   v1.34.8-eks-3385e9b
192.168.20.251   100.31.45.28      Amazon Linux 2023.11.20260526   6.12.88-
119.157.amzn2023.x86_64   containerd://2.2.3+unknown
ip-192-168-24-56.ec2.internal    Ready    <none>   49m   v1.34.8-eks-3385e9b
192.168.24.56    3.85.222.54       Amazon Linux 2023.11.20260526   6.12.88-
119.157.amzn2023.x86_64   containerd://2.2.3+unknown
ip-192-168-40-151.ec2.internal   Ready    <none>   48m   v1.34.8-eks-3385e9b
192.168.40.151   98.93.191.21      Amazon Linux 2023.11.20260526   6.12.88-
119.157.amzn2023.x86_64   containerd://2.2.3+unknown
ip-192-168-41-174.ec2.internal   Ready    <none>   48m   v1.34.8-eks-3385e9b
192.168.41.174   13.221.147.229    Amazon Linux 2023.11.20260526   6.12.88-
119.157.amzn2023.x86_64   containerd://2.2.3+unknown
```

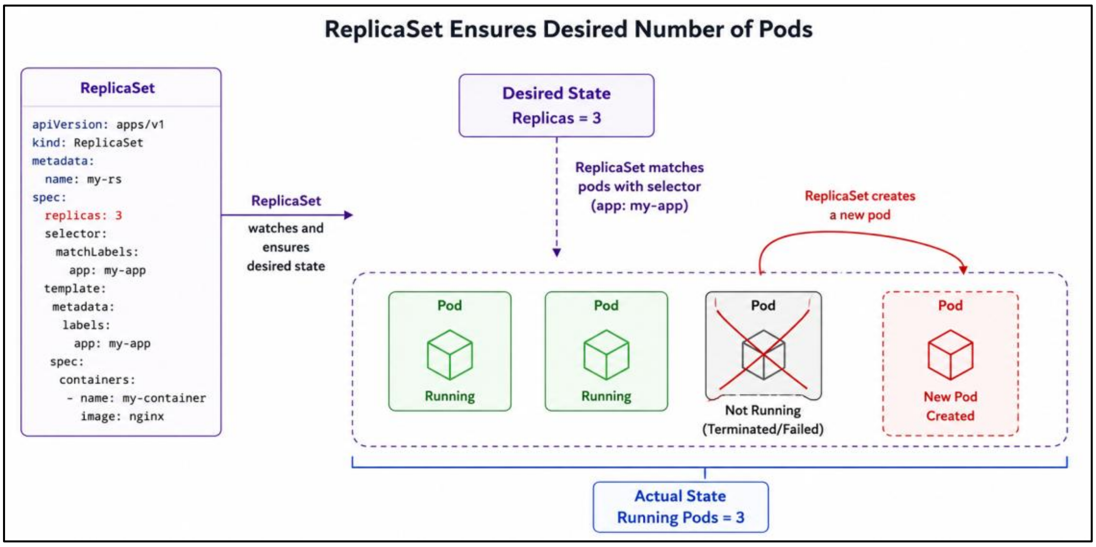
```
ip-192-168-7-37.ec2.internal     Ready    <none>   49m   v1.34.8-eks-3385e9b
192.168.7.37     98.82.116.227    Amazon Linux 2023.11.20260526   6.12.88-
119.157.amzn2023.x86_64   containerd://2.2.3+unknown

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods -o wide
NAME                  READY   STATUS    RESTARTS   AGE   IP                NODE
NOMINATED NODE   READINESS GATES
fb-pod                1/1     Running   0          33m   192.168.36.105    ip-
192-168-40-151.ec2.internal   <none>           <none>
pod1                  1/1     Running   0          44m   192.168.25.228    ip-
192-168-20-251.ec2.internal   <none>           <none>
two-container-pod     2/2     Running   0          19m   192.168.58.252    ip-
192-168-41-174.ec2.internal   <none>           <none>

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl drain ip-192-168-40-151.ec2.internal --ignore-daemonsets --force
node/ip-192-168-40-151.ec2.internal cordoned
Warning: deleting Pods that declare no controller: default/fb-pod; ignoring
DaemonSet-managed Pods: kube-system/aws-node-pdwld, kube-system/kube-proxy-
jt8fd
evicting pod default/fb-pod
pod/fb-pod evicted
node/ip-192-168-40-151.ec2.internal drained

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods -o wide
NAME                  READY   STATUS    RESTARTS   AGE   IP                NODE
NOMINATED NODE   READINESS GATES
pod1                  1/1     Running   0          46m   192.168.25.228    ip-
192-168-20-251.ec2.internal   <none>           <none>
two-container-pod     2/2     Running   0          21m   192.168.58.252    ip-
192-168-41-174.ec2.internal   <none>           <none>

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get nodes
NAME                             STATUS                     ROLES    AGE
VERSION
ip-192-168-20-251.ec2.internal   Ready                      <none>   53m
v1.34.8-eks-3385e9b
ip-192-168-24-56.ec2.internal    Ready                      <none>   53m
v1.34.8-eks-3385e9b
ip-192-168-40-151.ec2.internal   Ready,SchedulingDisabled   <none>   53m
v1.34.8-eks-3385e9b
ip-192-168-41-174.ec2.internal   Ready                      <none>   53m
v1.34.8-eks-3385e9b
ip-192-168-7-37.ec2.internal     Ready                      <none>   53m
v1.34.8-eks-3385e9b

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods -o wide
NAME                  READY   STATUS    RESTARTS   AGE   IP                NODE
NOMINATED NODE   READINESS GATES
pod1                  1/1     Running   0          48m   192.168.25.228    ip-
192-168-20-251.ec2.internal   <none>           <none>
two-container-pod     2/2     Running   0          23m   192.168.58.252    ip-
192-168-41-174.ec2.internal   <none>           <none>

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl delete pod pod1
pod "pod1" deleted from default namespace
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP               NODE
NOMINATED NODE   READINESS GATES
two-container-pod   2/2     Running   0          24m   192.168.58.252   ip-192-168-41-174.ec2.internal   <none>           <none>

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl uncordon ip-192-168-40-151.ec2.internal
node/ip-192-168-40-151.ec2.internal uncordoned

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get nodes
NAME                             STATUS   ROLES    AGE   VERSION
ip-192-168-20-251.ec2.internal   Ready    <none>   62m   v1.34.8-eks-3385e9b
ip-192-168-24-56.ec2.internal    Ready    <none>   62m   v1.34.8-eks-3385e9b
ip-192-168-40-151.ec2.internal   Ready    <none>   62m   v1.34.8-eks-3385e9b
ip-192-168-41-174.ec2.internal   Ready    <none>   62m   v1.34.8-eks-3385e9b
ip-192-168-7-37.ec2.internal     Ready    <none>   62m   v1.34.8-eks-3385e9b
```

Replica Set

- if one Pod got deleted, Replica set object creates another Pod

- Replica Set object maintains desired number of pods/pod’s replicas always up and running.



```yaml
#4_rs.yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: instagram-replicaset
  labels:
    app: instagram-rs # replicaset label
spec:
  replicas: 3
  selector:
    matchLabels:
      app: instagram-app # this is the label selector, it will select the pods with this label and create 3 replicas of those pods.
  template:
    metadata:
      labels:
        app: instagram-app # this is the label of the pod, we can use this label to select the pod.
    spec:
      containers:
      - name: instagram-container
        image: tomcat
        ports:
        - containerPort: 8080
```

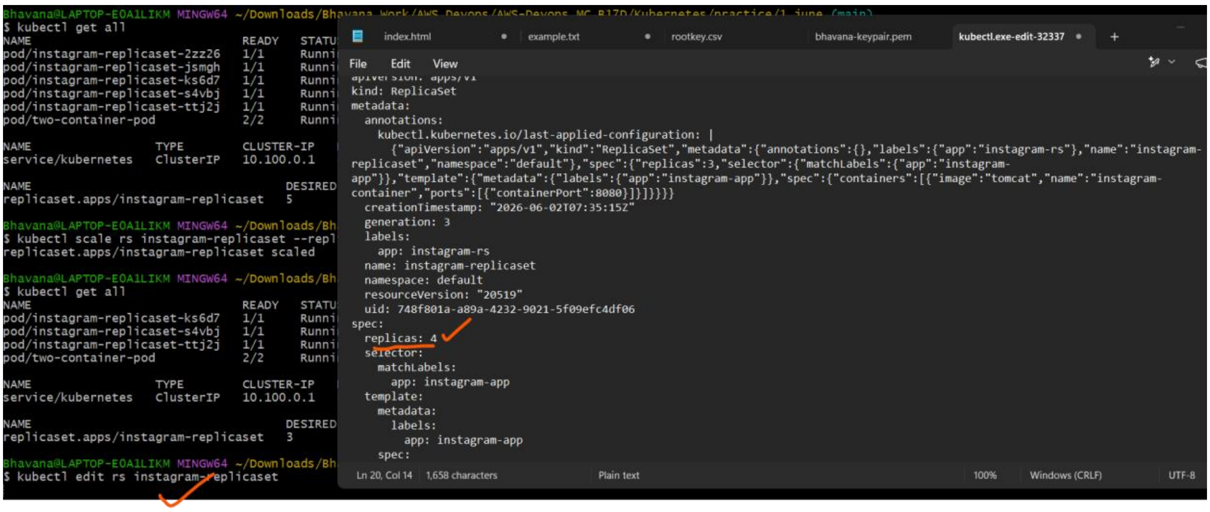
```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl apply -f 4_rs.yaml
replicaset.apps/instagram-replicaset created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods
NAME                         READY   STATUS    RESTARTS      AGE
instagram-replicaset-7xn9v   1/1     Running   0             9s
instagram-replicaset-ks6d7   1/1     Running   0             9s
instagram-replicaset-ttj2j   1/1     Running   0             9s
two-container-pod            2/2     Running   1 (86s ago)   61m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME                             READY   STATUS    RESTARTS       AGE
pod/instagram-replicaset-7xn9v   1/1     Running   0              29s
pod/instagram-replicaset-ks6d7   1/1     Running   0              29s
pod/instagram-replicaset-ttj2j   1/1     Running   0              29s
pod/two-container-pod            2/2     Running   1 (106s ago)   61m

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   99m

NAME                                   DESIRED   CURRENT   READY   AGE
replicaset.apps/instagram-replicaset   3         3         3       31s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get rs
NAME                   DESIRED   CURRENT   READY   AGE
instagram-replicaset   3         3         3       100s
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods
NAME                          READY   STATUS    RESTARTS       AGE
instagram-replicaset-7xn9v    1/1     Running   0              112s
instagram-replicaset-ks6d7    1/1     Running   0              112s
instagram-replicaset-ttj2j    1/1     Running   0              112s
two-container-pod             2/2     Running   1 (3m9s ago)   63m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl delete pod instagram-replicaset-7xn9v
pod "instagram-replicaset-7xn9v" deleted from default namespace

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods
NAME                          READY   STATUS    RESTARTS        AGE
instagram-replicaset-ks6d7    1/1     Running   0               2m17s
instagram-replicaset-s4vbj    1/1     Running   0               6s
instagram-replicaset-ttj2j    1/1     Running   0               2m17s
two-container-pod             2/2     Running   1 (3m34s ago)   63m


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl describe rs instagram-replicaset
Name:         instagram-replicaset
Namespace:    default
Selector:     app=instagram-app
Labels:       app=instagram-rs
Annotations:  <none>
Replicas:     3 current / 3 desired
Pods Status:  3 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=instagram-app
  Containers:
   instagram-container:
    Image:         tomcat
    Port:          8080/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Events:
  Type    Reason            Age    From                   Message
  ----    ------            ----   ----                   -------
  Normal  SuccessfulCreate  4m15s  replicaset-controller  Created pod: instagram-replicaset-
ttj2j
  Normal  SuccessfulCreate  4m15s  replicaset-controller  Created pod: instagram-replicaset-
ks6d7
  Normal  SuccessfulCreate  4m15s  replicaset-controller  Created pod: instagram-replicaset-
7xn9v
  Normal  SuccessfulCreate  2m4s   replicaset-controller  Created pod: instagram-replicaset-
s4vbj
```

- One pod from above was deleted and RS object created new Pod.

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get pods -o wide
NAME                          READY   STATUS    RESTARTS        AGE     IP
NODE                                    NOMINATED NODE   READINESS GATES
instagram-replicaset-ks6d7    1/1     Running   0               5m29s
192.168.36.105   ip-192-168-40-151.ec2.internal   <none>           <none>
instagram-replicaset-s4vbj    1/1     Running   0               3m18s
192.168.50.18    ip-192-168-40-151.ec2.internal   <none>           <none>
instagram-replicaset-ttj2j    1/1     Running   0               5m29s
192.168.25.228   ip-192-168-20-251.ec2.internal   <none>           <none>
two-container-pod             2/2     Running   1 (6m46s ago)   66m
192.168.58.252   ip-192-168-41-174.ec2.internal   <none>           <none>
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl scale rs instagram-replicaset --replicas 5
replicaset.apps/instagram-replicaset scaled

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME                               READY   STATUS    RESTARTS        AGE
pod/instagram-replicaset-2zz26     1/1     Running   0               10s
pod/instagram-replicaset-jsmgh     1/1     Running   0               10s
pod/instagram-replicaset-ks6d7     1/1     Running   0               7m47s
pod/instagram-replicaset-s4vbj     1/1     Running   0               5m36s
pod/instagram-replicaset-ttj2j     1/1     Running   0               7m47s
pod/two-container-pod              2/2     Running   1 (9m4s ago)    69m

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   106m

NAME                                   DESIRED   CURRENT   READY   AGE
replicaset.apps/instagram-replicaset   5         5         5       7m49s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl scale rs instagram-replicaset --replicas 3
replicaset.apps/instagram-replicaset scaled

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME                               READY   STATUS    RESTARTS         AGE
pod/instagram-replicaset-ks6d7     1/1     Running   0                8m22s
pod/instagram-replicaset-s4vbj     1/1     Running   0                6m11s
pod/instagram-replicaset-ttj2j     1/1     Running   0                8m22s
pod/two-container-pod              2/2     Running   1 (9m39s ago)    69m

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   107m

NAME                                   DESIRED   CURRENT   READY   AGE
replicaset.apps/instagram-replicaset   3         3         3       8m24s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl edit rs instagram-replicaset
replicaset.apps/instagram-replicaset edited
```



```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME                              READY   STATUS    RESTARTS      AGE
pod/instagram-replicaset-ks6d7    1/1     Running   0             10m
pod/instagram-replicaset-rxlxp    1/1     Running   0             9s
pod/instagram-replicaset-s4vbj    1/1     Running   0             8m24s
pod/instagram-replicaset-ttj2j    1/1     Running   0             10m
pod/two-container-pod             2/2     Running   1 (11m ago)   72m

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   109m

NAME                                   DESIRED   CURRENT   READY   AGE
replicaset.apps/instagram-replicaset   4         4         4       10m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl delete rs instagram-replicaset
replicaset.apps "instagram-replicaset" deleted from default namespace

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME                    READY   STATUS    RESTARTS      AGE
pod/two-container-pod   2/2     Running   1 (14m ago)   75m

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   112m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl delete pod two-container-pod
pod "two-container-pod" deleted from default namespace

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get po
NAME        READY   STATUS    RESTARTS   AGE
insta-pod   1/1     Running   0          88s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl delete pod insta-pod
pod "insta-pod" deleted from default namespace


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   127m
```

```yaml
#5_pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: insta-pod
  labels:
    app: instagram-app
spec:
  containers:
  - name: instagram-container
    image: tomcat
    ports:
    - containerPort: 8080
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl apply -f 5_pod.yaml
pod/insta-pod created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME            READY   STATUS    RESTARTS   AGE
pod/insta-pod   1/1     Running   0          6s

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   127m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl apply -f 4_rs.yaml
replicaset.apps/instagram-replicaset created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME                             READY   STATUS    RESTARTS   AGE
pod/insta-pod                    1/1     Running   0          28s
pod/instagram-replicaset-mkp8m   1/1     Running   0          6s
pod/instagram-replicaset-nvp6l   1/1     Running   0          6s

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   128m

NAME                                   DESIRED   CURRENT   READY   AGE
replicaset.apps/instagram-replicaset   3         3         3       7s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get po --show-labels
NAME                         READY   STATUS    RESTARTS   AGE   LABELS
insta-pod                    1/1     Running   0          44s   app=instagram-app
instagram-replicaset-mkp8m   1/1     Running   0          22s   app=instagram-app
instagram-replicaset-nvp6l   1/1     Running   0          22s   app=instagram-app
```